Namespace LlamaLogic.Packages

Classes

ArrayBufferWriterOfByteStream

Represents a write-only Stream that collects what is written to it in memory (🔓)

DataBasePackedFile

Represents a Maxis DBPF format package file (🔒)

ReadOnlyMemoryOfByteStream

Represents a read-only, seekable Stream over a ReadOnlyMemory<T> of bytes (🔓)

ResourceFileTypeAttribute

Specifies the file type of a package resource type independent of storage within a DataBasePackedFile

ResourceFileTypeFormatAttribute

Specifies the extension and mime type of a file type of a package resource independent of storage within a DataBasePackedFile

ResourceToolingMetadataAttribute

Specifies that the ResourceType is tooling metadata and not intended for use by the game

SmartSimUtilities

Provides utility methods specific to The Sims 4

Structs

ResourceKey

Represents the unique key of a resource within a DataBasePackedFile

Enums

CompressionMode

Represents the compression mode to use when setting the content of a resource in a DataBasePackedFile

ResourceFileType

The file type of a package resource independent of storage within a DataBasePackedFile

ResourceKeyOrder

Specifies the order in which a ResourceKey sequence should be processed

ResourceType

A type of resource within a DataBasePackedFile

Namespace LlamaLogic.Architecture

Classes

Blueprint
Represents a tray blueprint (a `.blueprint` file): a lot's objects and architecture (🔓)

BlueprintImage
Represents an image stored in a Blueprint (🔓)

BlueprintObjectDefinitionMapping
Represents a pair of object definition ids stored in a Blueprint (🔓)

FloorFeatureLayer
Represents a layer of the FloorFeaturesSection: one type of floor feature, with a set of values per level

FloorFeatureLevel
Represents the values of a FloorFeatureLayer on one level

FloorFeaturesSection
Represents the FloorFeatures section: the floor feature layers (🔓)

FloorIdGridSection
Represents one of the seven floor id grids: the FloorKeysSection id painted on each quarter of each tile (🔓)

FloorKeyEntry
Represents an entry of the FloorKeysSection: a floor pattern key and the id the floor id grids use for it

FloorKeysSection
Represents the FloorKeys section: the floor pattern keys the floor id grids refer to by id (🔓)

FloorRotationGridSection
Represents one of the five floor rotation grids: the rotation of the pattern on each quarter of each tile of a FloorIdGridSection (🔓)

LegacyFencePoint
Represents an entry of the LegacyFencePointsSection

LegacyFencePointsSection
Represents the LegacyFencePoints section: two lists of points with fence and ceiling rail keys, found only in older world files (🔓)

LegacyWallGraphEdge
Represents an edge of Edges

LegacyWallGraphEdgePart
Represents a part of a LegacyWallGraphEdge

LegacyWallGraphProductEdge
Represents an edge of ProductEdges1 or ProductEdges2

LegacyWallGraphSection
Represents the LegacyWallGraph section: a graph of wall edges with their wall patterns and products, found only in older world files (🔓)

LotArchitecture
Represents the architecture of a lot: its walls, floors, roofs, stairs, terrain and the rest of its build, as typed sections (🔓)

LotArchitectureSection
Represents one section of a lot's architecture: one record of the lot's RecordBundle

LotInfoPlexValue
Represents one entry of the plex map of a LotInfoSection: an optional value per plex unit

LotInfoSection
Represents the LotInfo section: the lot's levels, style, blueprint and users (🔓)

LotInfoUser
Represents one of the two user records of a LotInfoSection

LotObjectsSection
Represents the LotObjects section: the objects of the lot, as an `EA.Sims4.Network.LotObjectList` message (🔓)

LotSizeSection
Represents the LotSize section: the lot's dimensions (🔓)

ModularPiece
Represents a modular piece of the ModularPiecesSection

ModularPiecesSection
Represents the ModularPieces section: the modular pieces placed on the lot (🔓)

ObjectFallbacksSection
Represents the ObjectFallbacks section: the game's object fallback data, four sets of object ids (🔓)

ObjectTileBounds
Represents one entry of an ObjectTileBoundsSection: the tiles an object covers

ObjectTileBoundsSection

Represents the ObjectTileBounds section: the tile rectangles of the lot's objects (🔓)

PendingPlacementSection

Represents the PendingPlacement section: a footprint the game places on the lot after loading (🔓)

Plex

Represents a plex (a unit of a multi-unit lot) of the PlexesSection, with the blocks that make up its outline

PlexGroup

Represents a group of the PlexGroupsSection

PlexGroupMap

Represents an entry of the map from which the game looks up groups of the PlexGroupsSection

PlexGroupMapEntry

Represents an entry of a PlexGroupMap

PlexGroupsSection

Represents the PlexGroups section: groups of values the PlexesSection refers to by id (🔓)

PlexesSection

Represents the Plexes section: the lot's plexes (units), with the blocks of their outlines (🔓)

Pond

Represents a pond of the PondsSection

PondCell

Represents a tile of a Pond and the quarters of it the pond covers

PondsSection

Represents the Ponds section: the lot's ponds and the tile quarters each one covers (🔓)

ProductTable

Represents a table of catalog product keys and the ids by which the rest of a section refers to them

ProductTableEntry

Represents an entry of a ProductTable: a catalog product key and its id

ResourceManifestSection

Represents the ResourceManifest section: the keys of the catalog resources the other sections reference (🔓)

RevisionSection

Represents the Revision section: the lot's revision counter (🔓)

Roof
Represents one roof of a RoofsSection

RoofsSection
Represents the Roofs section: the lot's roofs (🔓)

Stair
Represents one stair or ladder of a StairsSection

StairStage
Represents one stage of a Stair: a straight run of steps

StairsSection
Represents the Stairs section: the lot's stairs and ladders (🔓)

TerrainHeightsSection
Represents the TerrainHeights section: the lot's terrain height grid, a second vertex grid of unknown role and a water level per tile quarter (🔓)

TerrainPaintLayer
Represents a layer of the TerrainPaintSection: a terrain paint and the mask of where it is applied

TerrainPaintSection
Represents the TerrainPaint section: the terrain paint layers, each a paint and a mask of where it is applied (🔓)

Unknown7B33C1F5Section
Represents the Unknown7B33C1F5 section: two lists of 64-bit values whose meaning has not been established (🔓)

UnknownC9106475Entry
Represents one entry of an UnknownC9106475Section

UnknownC9106475Level
Represents one level of an UnknownC9106475Entry

UnknownC9106475Section
Represents the UnknownC9106475 section: a list of entries, each holding levels with keys, whose meaning has not been established (🔓)

VenueBusinessInfo
Represents one business entry of a VenueSection

VenueSection

Represents the Venue section: the lot's venue (🔓)

VertexGraph

Represents one of the graphs of vertices and edges of the VertexGraphsSection

VertexGraphEdge

Represents an edge of a VertexGraph

VertexGraphVertex

Represents a vertex of a VertexGraph

VertexGraphsSection

Represents the VertexGraphs section: graphs of vertices and edges, whose version decides the layout of the WallEdgesSection (🔓)

WallBlock

Represents a block: a room, platform, wall line, fence line, pool or fountain, as loops of segments on one level, with the catalog products along them

WallBlockLoop

Represents a closed or open polygon of a WallBlock, as its segments

WallBlockSegment

Represents a segment of a WallBlockLoop: the run from one point of the loop to the next, with the catalog products along it

WallBlocksSection

Represents the WallBlocks section: the lot's blocks, which are its rooms, wall lines, fence lines, pools and fountains (🔓)

WallEdgePatterns

Represents an entry of WallPatternEdges: the two wall patterns of a wall edge

WallEdgeProduct

Represents an entry of ProductEdges or AdditionalProductEdges: the catalog product along a wall edge

WallEdgesLegacyEntry

Represents an entry of LegacyEntries

WallEdgesSection

Represents the WallEdges section: the wall patterns and catalog products along the lot's wall edges (🔓)

WeatherSection

Represents the Weather section: the lot's seasonal weather interpolation value (🔓)

WorldLotObjectsModel

Represents a WorldLotObjects resource: the objects of a lot template in a world file, as a `LotObjectList` protobuf message in protobuf text format (🔓)

ZoneArchitecture

Provides access to the architecture of the lots of a zone in a save game: the `architecture_data` field of the `EA.Sims4.Persistence.ZoneObjectData` protobuf message, the content of a ZoneObjectData resource

Structs

RoofPoint

Represents a lot-local tile position of a Roof's footprint

Enums

FloorQuarter

The four triangular quarters of a tile, named by the side of the tile each one touches, in the order the game stores them

LotArchitectureSectionType

The types of the sections of a lot's architecture, each of which is one record of the lot's RecordBundle

WallBlockSegmentKind

Specifies the kind of a WallBlockSegment

WallBlockType

Specifies the type of a WallBlock

Namespace LlamaLogic.TuningDescriptions

Classes

EnumTunableDescription

Represents the description of a tunable whose value is a member of an enumeration (a `TunableEnum` element) (🔒)

FragmentReferenceTunableDescription

Represents a reference to a shared description (a `TdescFragTag` element), whose Class names a TuningDescriptionFragment (🔒)

ListTunableDescription

Represents the description of a tunable that is a sequence of values described by Item (a `TunableList` element), including sets and mappings (🔒)

ScalarTunableDescription

Represents the description of a single-valued tunable (a `Tunable` element) (🔒)

SimDataGenerationException

Represents an exception thrown by SimDataGenerator when it cannot generate SimData for tuning

SimDataGenerator

Generates SimData resources from tuning the way Maxis' build does, using the tuning descriptions of the game version for which they are generated (🔒)

TunableDescription

Represents the description of a tunable (🔒)

TuningDescriptionAction

Represents a change Maxis recorded in a document's version history, identifying tunables by their Muid (🔒)

TuningDescriptionClass

Represents a class described in a TuningDescriptionModule (a `Class` element): an enumeration, the holder of module tuning, or both (🔒)

TuningDescriptionDocument

Represents a document of a TuningDescriptionSet: an instance, a module, or a fragment (🔒)

TuningDescriptionEnumItem

Represents a member of an enumeration described by a TuningDescriptionClass (an `EnumItem` element) (🔒)

TuningDescriptionFragment
Represents a description shared by the FragmentReferenceTunableDescriptions that name it (a `TdescFrag` element), such as `TunableTestSet` (🔒)

TuningDescriptionInstance
Represents the description of a class of instance tuning (an `Instance` element), such as `Buff` (🔒)

TuningDescriptionModule
Represents the description of a Python module's tuning and enumerations (a `Module` element), such as `sims.sim_info_types` (🔒)

TuningDescriptionSet
Represents a set of tuning descriptions for a version of The Sims 4, as Maxis distributes it (🔒)

TupleTunableDescription
Represents the description of a tunable composed of named tunables (a `TunableTuple` element) (🔒)

UnknownTunableDescription
Represents a description in an element this library does not recognize, kept so that nothing in a document is lost (🔒)

VariantTunableDescription
Represents the description of a tunable whose value is one of several named options (a `TunableVariant` element) (🔒)

Enums

TuningDescriptionActionType
The type of a TuningDescriptionAction

TuningDescriptionExportModes
The destinations to which the game's build exports a tunable (its `export_modes` attribute)

Namespace LlamaLogic.TuningArchive

Classes

GameInstallation
: Represents an installation of The Sims 4 from which a TuningArchiveFile can capture a snapshot (🔒)

GameInstallationPack
: Represents the base game or a pack in a GameInstallation (🔒)

GameInstallationPackage
: Represents a package file of a GameInstallationPack and the priority the game loads it with

Sims4StudioBatchImport
: Names files the way Sims 4 Studio's batch import reads them

TuningAnalysis
: Represents a tuning document understood through a set of tuning descriptions

TuningArchiveCaptureOptions
: Represents the choices a creator makes when capturing a snapshot from an installation of the game

TuningArchiveCaptureResult
: Represents the outcome of capturing or importing a snapshot into a TuningArchiveFile

TuningArchiveChange
: Represents how a resource differs, if at all, between two snapshots

TuningArchiveDescriptions
: Represents a set of tuning descriptions (TDESCs) imported into a TuningArchiveFile

TuningArchiveEntry
: Represents a version of a resource in a TuningArchiveFile

TuningArchiveExportOptions
: Represents the choices a creator makes when exporting a snapshot as files, the way Scumbumbo's XML Extractor extracts the game's tuning

TuningArchiveExportResult
: Represents the outcome of exporting a snapshot as files

TuningArchiveFile
: Represents an archive of The Sims 4's tuning across versions of the game, stored in a SQLite database file (🔒)

TuningArchiveHistoryItem

Represents a version of a resource and the snapshots that contain it

TuningArchiveImportOptions

Represents the choices a creator makes when importing a folder of tuning files extracted by another tool as a snapshot

TuningArchiveReference

Represents a reference from a tuning to another tuning or resource

TuningArchiveSnapshot

Represents the state of the game's tuning at a point in time, as captured in a TuningArchiveFile

TuningArchiveSnapshotPack

Represents a pack that was present when a TuningArchiveSnapshot was taken

TuningArchiveStringChange

Represents how a string differs, if at all, between two snapshots

TuningArchiveUninstalledTuning

Represents a tuning of a pack that was not installed when a TuningArchiveSnapshot was captured, known only from the pack's patches in the game's `Delta` folder

TuningCommenter

Adds to tuning the comments XML Extractor adds, as a snapshot knows them: the text of each string after its key, and the class and name of the tuning or animation state machine each reference refers to

TuningCoreLibraryDraft

Drafts a Lot 51 Core Library tuning injector snippet (for Core Library 1.43) that makes changes to tuning without overriding it

TuningDescriptionCache

Keeps the documents of a TuningDescriptionSet that analysis has used, so that analyzing many tunings does not read and parse the same descriptions repeatedly (🔒)

TuningDraftDiagnostic

Represents what a draft does, or cannot do, about a change to a tunable

TuningDraftResult

Represents a draft that makes changes to tuning without overriding it: a script or a snippet

TuningDraftTarget

Represents a tuning a creator has changed in a draft: the tuning as the archive has it and as the creator wants it

TuningFolderUpdate

Finds the overrides of game tuning in every package in a folder and its subfolders, works out what a game version means for each, checks the SimData of the tuning the packages add against the game version's tuning descriptions, and writes updated copies of the packages with the changes that need no judgment from the creator, with a report of everything

TuningFolderUpdateOptions

Represents which changes a folder update makes beyond the merges that need no judgment from the creator

TuningFolderUpdatePackage

Represents a package in a folder being updated

TuningFolderUpdateReportRow

Represents a row of a folder update's report: an override, the SimData of a tuning a package adds, or a package that couldn't be read

TuningFolderUpdateResult

Represents what writing a folder update did

TuningFolderUpdateWrite

Represents what writing a folder update did for one of its packages

TuningMerge

Merges a mod's changes to a tuning onto the game's new version of it: a three-way merge of the version the mod was made from (the base), the mod's version, and the game's new version

TuningMergeConflict

Represents a place where a merge could not combine the mod's change with the game's

TuningMergeResult

Represents the result of a merge of a mod's changes onto the game's new version of a tuning

TuningNode

Represents an element of a tuning document, as understood through its tuning description

TuningPackageAddedSimData

Represents the SimData of tuning a package adds (rather than overrides), checked against SimData generated from the tuning with the tuning descriptions of the game version the package is being updated for

TuningPackageAddedSimDataPlan

Represents what a set of fixes does to the SimData of tuning a package adds, and to the tuning

TuningPackageOverride
: Represents a mod's override of a game tuning, and what a game version means for it

TuningPackageUpdate
: Finds a mod package's overrides of game tuning, works out what a game version means for each, and writes an updated copy of the package

TuningPackageUpdateChange
: Represents what writing an updated package did to a resource

TuningPythonDraft
: Drafts a Python script mod that makes changes to tuning as the game loads it, without overriding any tuning

TuningSemanticChange
: Represents an element that differs between two versions of a tuning document

TuningSemanticDiff
: Compares two versions of a tuning document element by element, matching elements by name and aligning list items, rather than line by line

TuningSimDataDraftResult
: Represents the SimData an override of a tuning needs

TuningSimDataDrafter
: Generates the SimData that overrides of a snapshot's tuning need, trusting the result only when the generator reproduces the game's own SimData for the original tuning

TuningXml
: Provides the canonical text form in which a tuning archive stores tuning and SimData documents

TuningXmlInjectorDraft
: Drafts an XML Injector snippet (for Scumbumbo's XML Injector, version 4) that makes changes to tuning without overriding it

Structs

TuningArchiveProgress
: Represents the progress of a long-running TuningArchiveFile operation

Enums

TuningArchiveChangeKind
: Specifies how a resource differs, if at all, between two snapshots

TuningArchiveExportAttributeOrder

Specifies the order in which ExportAsync(TuningArchiveSnapshot, string, TuningArchiveExportOptions?, IProgress<TuningArchiveProgress>?, CancellationToken) writes the `n` and `t` attributes of an element that has both

TuningArchiveExportExistingFiles

Specifies what ExportAsync(TuningArchiveSnapshot, string, TuningArchiveExportOptions?, IProgress<TuningArchiveProgress>?, CancellationToken) does when a file it would write already exists

TuningArchiveExportNaming

Specifies how ExportAsync(TuningArchiveSnapshot, string, TuningArchiveExportOptions?, IProgress<TuningArchiveProgress>?, CancellationToken) names the files it writes, following the conventions of Scumbumbo's XML Extractor

TuningArchiveResourceKind

Specifies the kind of a resource in a TuningArchiveFile

TuningArchiveSnapshotSource

Specifies where the contents of a TuningArchiveSnapshot came from

TuningDraftSeverity

Specifies what a draft can do about a change

TuningMergeChoice

Specifies which side of a merge conflict a merge takes

TuningMergeConflictKind

Specifies why a merge could not combine the mod's change with the game's

TuningPackageAddedSimDataFixes

Specifies which fixes writing an updated package makes to the SimData of tuning it adds, and to that tuning

TuningPackageAddedSimDataProblems

Specifies what is wrong with the SimData of tuning a package adds, by the tuning descriptions of the game version it is being updated for

TuningPackageOverrideOutcome

Specifies what a game version means for a mod's override of a tuning

TuningPackageSimDataChanges

Specifies which changes writing an updated package makes to its SimData

TuningPackageSimDataState

Specifies whether a mod's SimData for an override is what the override's tuning needs in the game's new version

TuningPackageUpdateChangeKind

Specifies what writing an updated package did to a resource

TuningSemanticChangeKind

Specifies how an element differs between two versions of a tuning document

TuningSimDataDraftKind

Specifies what an override of a tuning needs from its SimData

TuningValueKind

Specifies what the value of an element of a tuning document means, according to its tuning description

Namespace LlamaLogic.ValveDataFormat

Classes

VdfKeyValuePair

Represents a key-value pair in a VDF document

VdfNode

Represents a node in a VDF document

VdfSection

Represents a section in a VDF document